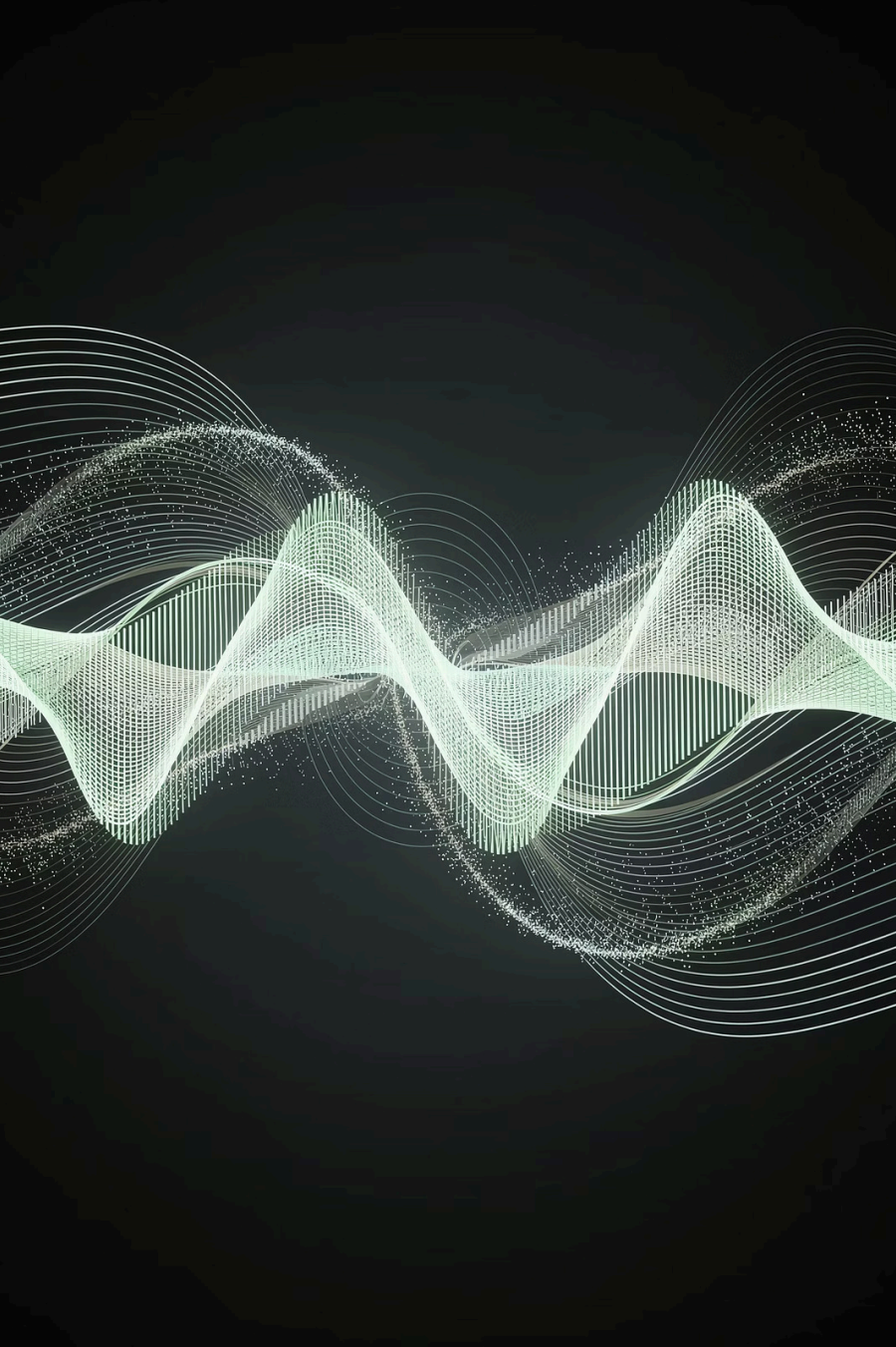




COM-304 • FOUNDATION MODELS • SPRING 2026

Nano4M-Audio:

Adding Audio as a 5th Modality to the 4M architecture.

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Can a Shared Transformer Absorb Audio at a small scale?



Audio: Dog Barking



Shared Transformer



RGB: Dog Image

Narrowing the problem and Self-Building the Dataset

We narrowed the task to animal sounds because audio-to-image alignment works best when the sound is strongly and unambiguously tied to a visual subject. Animal vocalizations like a dog barking, a cat meowing, or a cow lowing have a clear one-to-one mapping to a visual category.

That makes evaluation clean and interpretable: if the model learns cross-modal alignment, it should be able to go from audio to the correct animal image.

01

PANN

Audio-Caption Similarity

Pretrained Audio Neural Network that scores how well an audio clip matches a caption like “a dog barking.” This removes clips where the sound does not match the expected animal audio.

02

CLIP

Frame-Caption Similarity

Compares a video frame with the same caption to verify that the visual content matches the animal class label.

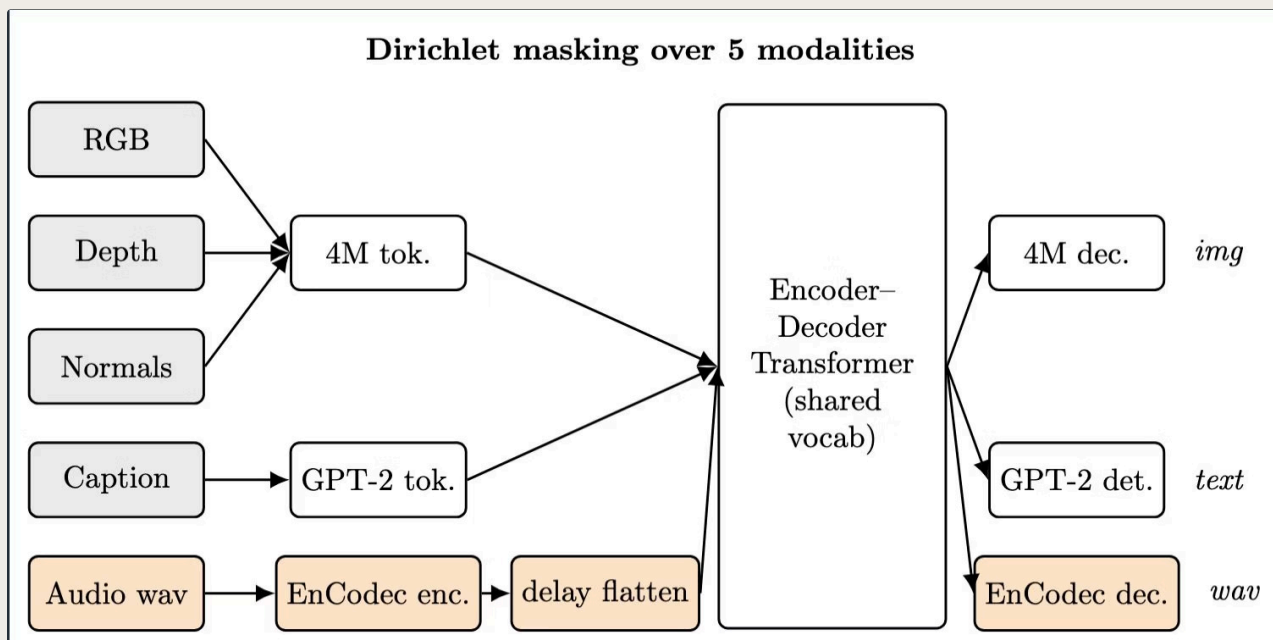
03

VAD

Voice Activity Detection

Filters out clips with human speech in the background so the final audio remains clean animal sound rather than narration or commentary.

Architecture



Tokenization per Modality

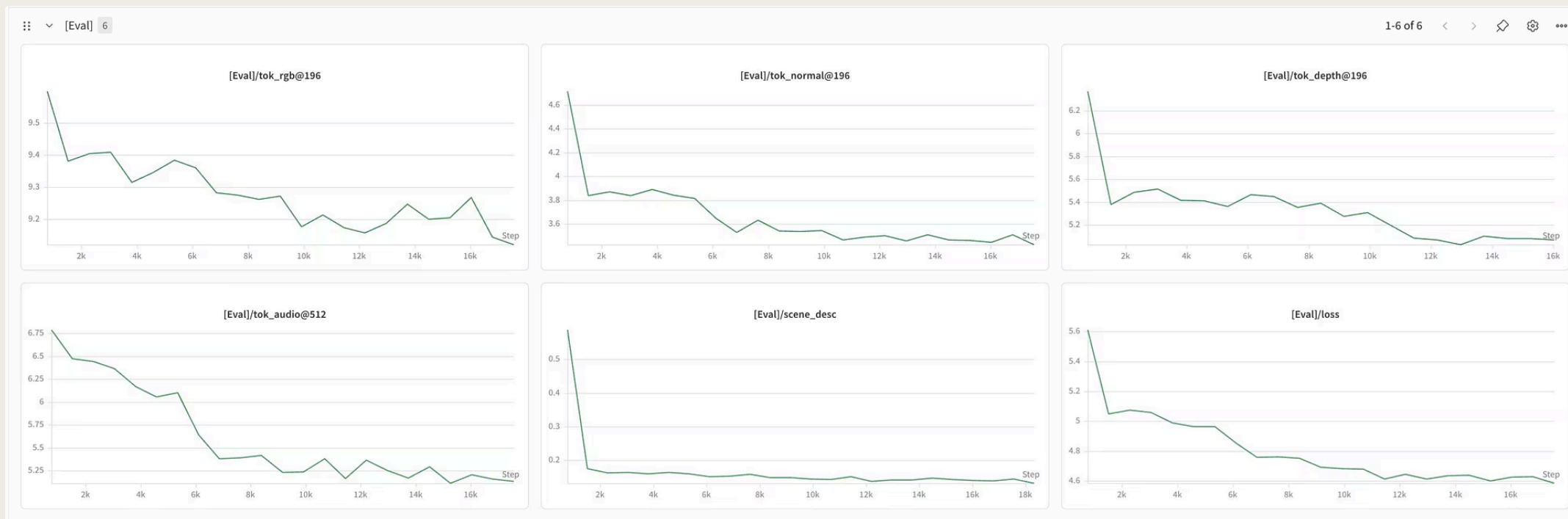
- **Audio** → EnCodec 24 kHz @ 1.5 kbps, K=2 RVQ, MusicGen delay-flatten → **512 tokens / 3.4 s**
- **RGB / Depth / Normal** → existing 4M tokenizers (16k / 8k / 8k vocab)
- **Caption** → GPT-2 BPE tokenizer

Architecture

- Encoder-decoder transformer: **d6-w512, ~86 M params** — unchanged by design
- Dirichlet masking; **span masking added for audio**, instead of the default random masking

Training

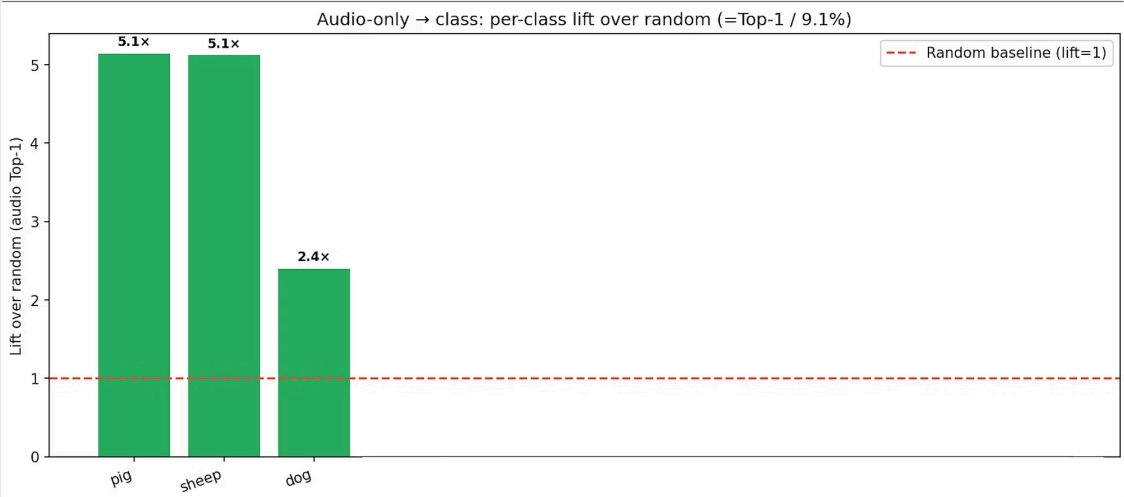
After many experimental runs and extensive tuning of hyperparameters and architectural choices — tokenizer selection, masking strategy, learning rate schedules, and modality weighting — the following represent our final run. This is the configuration we converged on.



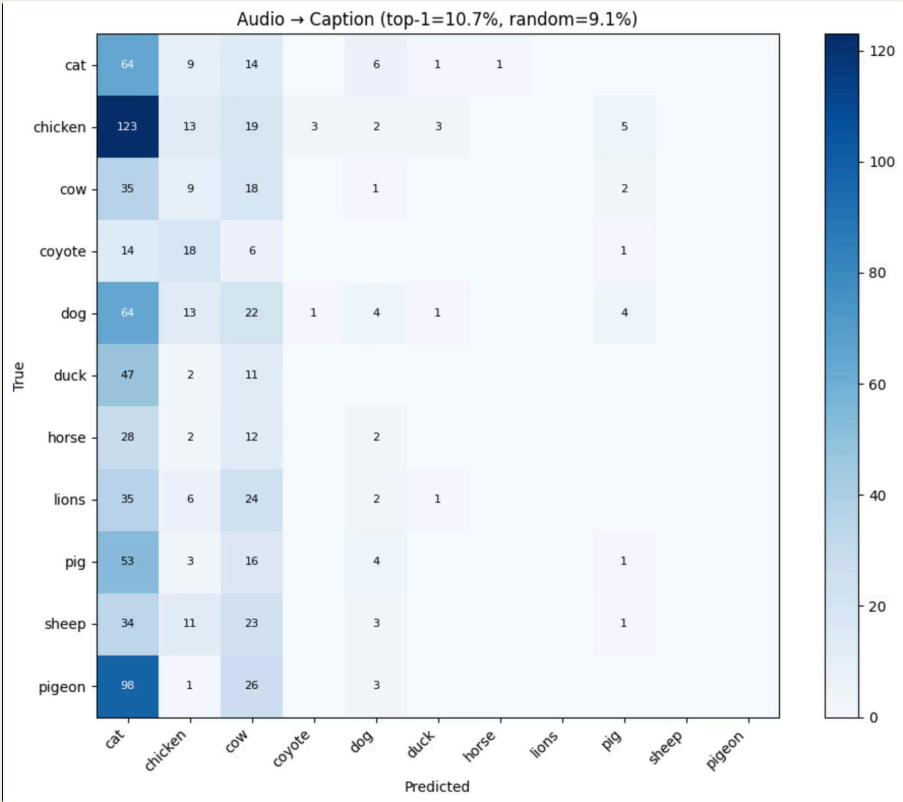
W&B training curves for all 5 modalities over 18k steps. Train and eval track tightly no overfitting detected.

Nano-4M understanding of the Audio modality.

The model achieves marginal lift over random overall, but the signal is concentrated in just 3 of 11 classes.



Per-class lift over random (Top-1 / 9.1%). Pig and sheep at 5.1x, dog at 2.4x – 8 other classes collapse to zero.



Audio → Caption confusion matrix (top-1=10.7%, random=9.1%).

Cross-Modal Generation: Where nano4M-Audio Falls Short

Latent Retrieval (Audio - Vision)

- 200 candidates · 4 directions: audio→RGB, RGB→audio, audio→Depth, Depth→audio
- Best R@5 = 4.5% vs. 2.5% random → 1.8× lift
- All other directions sit at or below random baseline

❌ Near-random retrieval across all directions — the latent space has no cross-modal structure.

Conditioned Generation (Audio → RGB)

- RGB images generated from audio conditioning are not recognizable as any animal class
- Confirmed by both an off-the-shelf image classifier and manual inspection

⚠️ The audio-vision bridge has not formed. The model cannot generate a meaningful image from an audio input.

Why does this happen?

Tokenizer Mismatch

EnCodec is an acoustic tokenizer optimized for reconstruction, not semantic alignment. Its tokens carry no class-level meaning.

Data Starvation

10,000 clips is a lot below the minimum needed for cross-modal structure to emerge at this parameter count.

Audio-conditioned cross-modal generation



Audio-conditioned cross-modal generation: RGB outputs conditioned on animal audio clips. Images are unrecognizable as the correct animal class — confirmed by classifier and manual inspection.

🔊 We will now listen to some of the audio clips generated by the model — judge for yourself whether the sound matches the intended animal class.

A Controlled Negative Result

What Worked

- Clean 5-modality pipeline end-to-end
- No overfitting; stable training across all modalities
- Span masking for audio integrates without issues
- Precise, isolatable diagnostic delivered

What Didn't

- Cross-modal alignment at this data scale
- Audio-conditioned generation quality
- Latent retrieval structure (audio - vision)

Path Forward

- Semantic audio tokenizer:
SpeechTokenizer or MERT over EnCodec
- $\geq 10\times$ data — full VGGSound (~150 k clips)